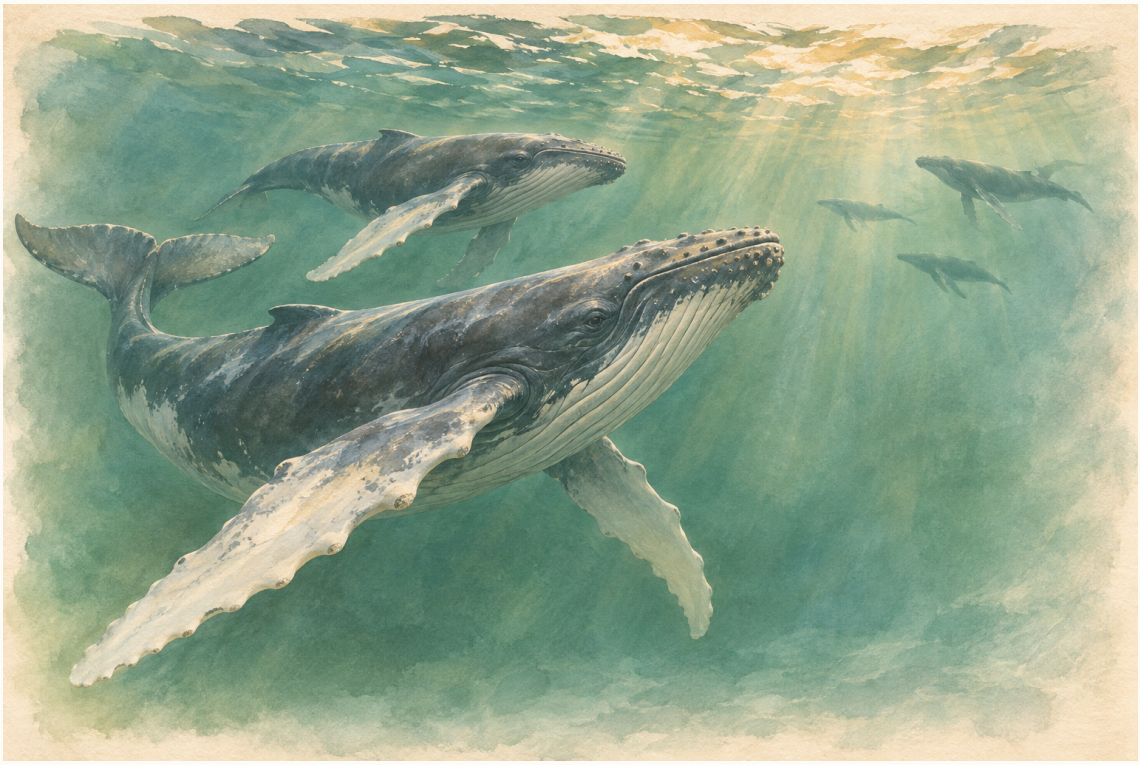
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A Change of Anthem in the Deep



新歌如何跨过海洋，传遍鲸群？

英文原文

Each winter, male humpback whales, and only males, fill their breeding grounds with a haunting melody. Sound units constitute phrases, and phrases build themes in an elaborate hierarchy. Each rendition lasts minutes, and the song can recur for hours, probably for courtship. All the males in a population sing an identical version, which evolves subtly each year.

That is the rule. Off eastern Australia, it was broken. From 1995 to 1998, researchers used hydrophones, or underwater microphones, to record humpbacks on their seasonal migration along the coastline. In the mid-1990s, merely two of 82 singers emitted a distinctive new song, utterly unlike the reigning chorus.

Within about two years, virtually every singer had converted, and by 1998 the old song had been displaced wholesale. A study published in Nature in November 2000 found that the new song resembled that of the Indian Ocean population off western Australia, apparently ferried east by a handful of visiting performers.

Nor was it a freak. In April 2011, a study in Current Biology analysed over a decade of recordings from six South Pacific populations. Song types primarily arose off eastern Australia, then rippled eastward through successive populations to French Polynesia, generally crossing the region within about two years.

Such rapid change, too fast for genetic evolution, points to imitation rather than inheritance, a persuasive case of cultural transmission. Why males sing, and why they overhaul their songs, remain controversial. One idea is that novelty itself spurs change. Some scholars are reluctant to define this as culture in the human sense. Whatever the label, a few outsiders off eastern Australia reshaped an entire population's music.

中文译文

每到冬季，雄性座头鲸——而且只有雄性——会让繁殖地回荡着令人难忘的旋律。声音单元组成乐句，乐句再构成主题，形成复杂精细的层级结构。每次歌唱持续数分钟，又能连续重复数小时，很可能是为了求偶。同一种群中的所有雄性都唱着相同版本，这个版本每年都会发生细微变化。

这是通常的规律，但在澳大利亚东部海域，规律被打破了。1995至1998年，研究者用水听器，也就是水下麦克风，记录沿岸季节性迁徙的座头鲸。20世纪90年代中期录到的82头歌者中，只有两头唱出了一首全新的歌，与当时占主流的合唱完全不同。

大约两年内，几乎每头歌者都换了歌；到1998年，旧歌已被彻底取代。2000年11月发表于《自然》的一项研究发现，新歌与澳大利亚西部近海的印度洋种群所唱的歌相似，似乎是少数来访歌者把它向东带来的。

这也并非孤例。2011年4月，《当代生物学》发表的一项研究分析了南太平洋六个种群十余年的录音。歌型大多起源于澳大利亚东部海域，随后依次经由一个个种群向东传播，直到法属波利尼西亚，通常约两年就能横跨这片区域。

这种变化快得难以用基因演化解释，指向的是模仿而非遗传，为文化传递提供了一个有说服力的例证。但雄性为何歌唱、为何全面更换歌曲，仍有争议。一种观点认为，新奇本身就会推动变化。有些学者不愿把它定义为人类意义上的文化。无论采用什么名称，澳大利亚东部海域的少数外来者，确实重塑了整个种群的音乐。

核心词汇

先记这22项。连着下方的原文短语一起回想。

constitute v.
构成
Sound units constitute phrases,

elaborate adj.
复杂精细的
themes in an elaborate hierarchy.

hierarchy n.
层级结构
themes in an elaborate hierarchy.

recur v.
反复出现
and the song can recur for hours,

courtship n.
求偶
for hours, probably for courtship.

identical adj.
完全相同的
population sing an identical version,

migration n.
迁徙
on their seasonal migration along the

distinctive adj.
与众不同的
emitted a distinctive new song,

virtually adv.
几乎
two years, virtually every singer had

displace v.
取代
old song had been displaced wholesale.

resemble v.
与……相似
the new song resembled that of the

a handful of n.phr.
少数；一小批
by a handful of visiting performers.

primarily adv.
主要地
Song types primarily arose off eastern

successive adj.
相继的
through successive populations to

evolution n.
演化
too fast for genetic evolution,

imitation n.
模仿
to imitation rather than inheritance,

inheritance n.
遗传
to imitation rather than inheritance,

transmission n.
传递；传播
case of cultural transmission.

overhaul v.
全面改造
why they overhaul their songs, remain

controversial adj.
存在争议的
their songs, remain controversial.

novelty n.
新奇
is that novelty itself spurs change.

reluctant adj.
不情愿的
Some scholars are reluctant to define

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

male adj.
雄性的

humpback whale n.phr.
座头鲸

breeding ground n.phr.
繁殖地

haunting adj.
萦绕心头的；难忘的

melody n.
旋律

sound unit n.phr.
声音单元

theme n.
主题；主题段落

rendition n.
一次演唱；演绎

population n.
生物种群

evolve v.
演变

subtly adv.
细微地；不易察觉地

off prep.
在……近海

researcher n.
研究者

microphone n.
麦克风

seasonal adj.
季节性的

coastline n.
海岸线

merely adv.
仅仅

emit v.
发出

utterly unlike adj.phr.
完全不同于

reigning adj.
当前占主导的

chorus n.
合唱

convert v.
转变；此处指改唱

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

wholesale adv.
全面地；全盘地

ferry v.
带来；运送

performer n.
表演者；此处指歌者

freak n.
罕见的例外

analyse v.
分析

recording n.
录音

Pacific proper n.
太平洋

ripple v.
如涟漪般扩散

eastward adv.
向东

Polynesia proper n.
波利尼西亚

generally adv.
通常

region n.
区域

rapid change n.phr.
快速变化

persuasive adj.
有说服力的

cultural adj.
文化的

spur v.
推动；促进

scholar n.
学者

define as phr.v.
与as连用：把……界定为

as prep.
标示被归入的类别

outsider n.
外来者

reshape v.
重塑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Sound units _____ phrases,

02 _____ every singer had converted,

03 _____ visiting performers.

04 Some scholars are _____ to define this as culture in the human sense.

文章主线

- 常态 同一种群的雄性座头鲸唱相同版本的歌，通常每年只作细微变化，歌唱可能用于求偶。
- 换歌 澳洲东部录到的82头鲸中，最初仅两头唱全新歌；约两年后，几乎整个种群都改唱新歌。
- 来源 新歌与澳洲西部种群的歌相似，研究推测是少量来访鲸把它带到了东部。
- 传播 六个南太平洋种群的十余年录音显示，歌型主要从澳洲东部逐群向东扩散，通常约两年传到法属波利尼西亚。
- 解释 变化太快，支持模仿而非遗传的解释；为何换歌、是否等同人类文化，仍有争议。

参考答案　1 constitute 2 virtually 3 a handful of 4 reluctant